

# NATIONAL TELE-OPHTHALMOLOGY AI SCREENING PROGRAM

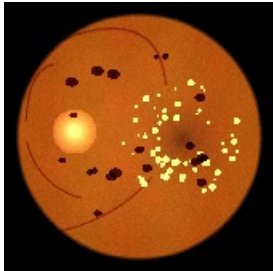
Explainable AI Clinical Decision Support System | Rural PHC Triage (SIH PS 26038)

**Patient ID:** IND-KA-PHC-8921    **Age/Sex:** 58 / Male    **Screening Date:** 15-Sep-2026 20:13  
**PHC Health Centre:** PHC Mulbagal, Kolar District    **Frontline Operator:** ASHA Worker #108  
**Eye Evaluated:** Right Eye (OD) (Nasal: Left)    **QA Usability Score:** 75/100 [PASS]

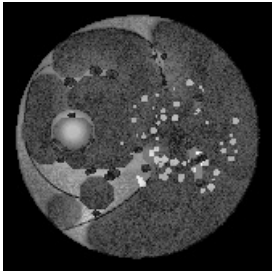
**TRIAGE ACTION: URGENT SPECIALIST REFERRAL REQUIRED (<48 HOURS)**

High risk of irreversible vision loss. Tertiary Vitreoretinal evaluation mandatory.

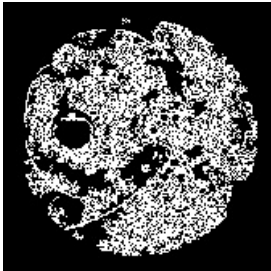
1. Raw Fundus (OD)



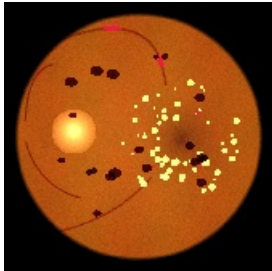
2. Enhanced Green



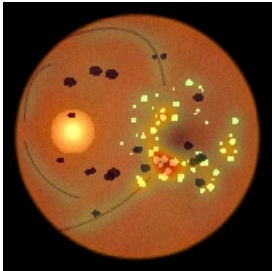
3. Retinal Vessels



4. Lesion Detections



5. Grad-CAM XAI

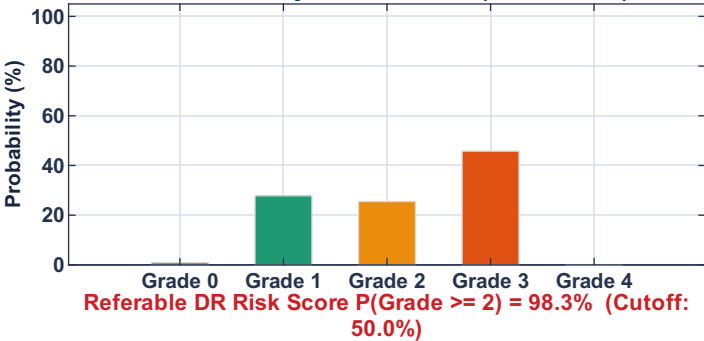


**Legend:** [Yellow]: Hard Exudates | [Ruby Red]: Microaneurysms & Hemorrhages | [Heatmap]: Deep Feature Attention (Grad-CAM)

## QUANTITATIVE BIOMARKERS & ETDRS 4-2-1 CLINICAL AUDIT

**ETDRS Staging:** MILD NPDR (Isolated microaneurysms only)  
**ETDRS 4-2-1 Rule:** 4Q Hemo: [1/4] | VB Quads: [0] | IRMA: [0]  
**Distance to Fovea (CSME):** None (Clear Macula) — Normal  
**Hard Exudates (Bright):** 0 clusters (0 px area)  
**Microaneurysms / Blot Hemo:** 9 MAs | 2 Hemorrhages  
**Vascular Density:** 50.34% FOV    **Optic Disc:** [103, 343] (Left)  
**Systemic EHR Profile:** T2D: 8 yrs | Last HbA1c: 8.2% | BP: 134 / 84 mmHg  
**Last Encounter Record:** 10-Nov-2025 (10 months ago) (Grade: Grade 1 (Mild NPDR))

## AI DR Severity: Severe NPDR (Conf: 45.8%)



## RECOMMENDED REFERRAL CARE: Minto Regional Institute of Ophthalmology (Govt. Medical College) [Level 3: Tertiary Medical College & Eye Institute]

**Est. Distance:** ~54.0 km | **Timeline:** HIGH PRIORITY REFERRAL WITHIN 14 DAYS | **Helpline:** +91 80 2670 1555 (Minto Triage Desk)

## OPHTHALMOLOGIST TELEMEDICINE REVIEW & VALIDATION (Target Review Time: under 30 Seconds)

Clinical Impression: [ ] Concordant with AI diagnosis [ ] Upgrade Severity [ ] Downgrade / Artefact  
Prescribed Action: [ ] Tele-Consultation Booked [ ] Anti-VEGF / Laser Referral [ ] PHC Annual Rescreen  
Reviewing Ophthalmologist Signature: ..... Date/Time: .....